

PERFORMANCE-BASED WORK STATEMENT (PWS)

CIRCUIT CARD ASSEMBLY BPA



**SOFTWARE
DIRECTORATE**

U.S AIR FORCE

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PWS Modification History

Date	Change	Paragraph #

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1 DESCRIPTION OF SERVICES

This requirement is to obtain electronic parts and products from industry and commercial Printed Circuit Boards (PCB) Fabrication and/or Assembly companies. This supports the 516th Common Aircraft Portable Reprogrammable Equipment (CAPRE), Common Built-in-Test Reprogramming Equipment and Portable Automated Test Station (CMBRE), Portable Automated Test Station (PATs) and similar missions. The requirement includes developing electronics solutions that solve Department of War (DoW) obsolescence issues and/or provides state-of-the-art hardware and software solutions to DoW Programs. This Performance Work Statement (PWS) will fill the need to (1) rapidly produce Printed Circuit Board Assemblies (PCBAs) for initial design and development use, (2) quickly produce prototype PCBAs for design validation, and (3) manufacture high-quality PCBAs for use in DoW operational environments.

2 SCOPE

The scope of this PWS is to enable the ordering of Printed Circuit Board Assemblies (PCBAs) through a turn-key solution for Printed Circuit Board (PCB) Fabrication and Assembly with multiple vendors. This is applicable to developmental, prototype, and production electronics systems for use by Department of War (DOW) Programs.

3 REFERENCE DOCUMENTS AND STANDARDS

3.1 Applicable Documents and Standards

The contractor shall comply with the following documents.

Document Number	Document Title	Date of Document
	PCB Design and Layout Files	As Provided
	PCB Fabrication Specification	As Provided

	Sub-component Bill of Materials (BOM)	As Provided
	PCB Assembly Design Files	As Provided
	PCB Assembly Specifications	As Provided

4 **BASIC SERVICES:** Turn Key Fabrication and Assembly services

- 4.1 General Requirements: The contractor shall produce/procure the products described within this section, based on the required specifications provided by the Government through individual BPA calls. The contractor shall have in-house capabilities for either PCB Fabrication or Assembly, or both. Sub-contracting for either capability is acceptable, but any sub-contractors utilized shall be approved by the Government POC in advance of the initiation of activities.
- 4.2 PCB Fabrication: The contractor shall produce requested quantities of PCBs in accordance with fabrication requirements and specification documents.
 - 4.2.1 The contractor shall fabricate PCBs in accordance with the specified IPC-A-600 classification; if not specified, the contractor shall fabricate PCBs to IPC Class 3 requirements.
 - 4.2.2 The contractor shall provide technical reports documenting electrical continuity test results, impedance matching test results, and/or results of Quality Control inspections. (CDRL A001, A002)
- 4.3 Component Procurement: The contractor shall procure sufficient materials and components, as identified on the Bill of Materials (BOM), to enable the assembly of requested quantities of PCBAs.
 - 4.3.1 The contractor shall only use authorized distributors or Original Equipment Manufacturers (OEMs) to source electronic components. Alternative sources may be used but must be approved by the Government POC.
 - 4.3.2 The contractor shall identify suitable alternates to components that are not immediately available or that have prohibitively long lead times. Any alternate components shall be approved by the Government POC.

- 4.4 PCB Assembly: The contractor shall assemble requested quantities of PCBAs in accordance with layout specifications and assembly requirement documents.
- 4.4.1 The contractor shall assemble PCBAs in accordance with specified J-STD-001 and IPC-A-610 classification; if not specified, the contract shall assemble PCBAs to IPC Class 3 requirements.
- 4.4.2 The contractor shall provide technical reports documenting the results of Quality Control inspections performed in accordance with specified IPC-A-610 inspections. (CDRL A003)
- 4.4.3 Unless otherwise specified, the contractor shall use Tin-Lead (Sn-Pb) solder.
- 4.4.4 The contractor shall return all excess materials and components to the Government after the completion of production activities.

5.0 SERVICES SUMMARY

PWS Paragraph	Performance Objective	Performance Threshold
4.1.2	Fabricate PCB's according to specifications	All PCB's fabricated to IPC Class 3 (or as specified) with documented evidence of circuit continuity.
4.3	Procure all necessary components for assembly.	All components from authorized distributors or other Government-approved sources.
4.4	Assembly PCBAs according to specifications.	All PCBAs assembly to IPC Class 2 (or as specified) with documented evidence of QC inspections.

6.0 CDRL Summary

6.1 CDRL Information: Contract data shall be delivered electronically IAW the corresponding requirement for each Data Item listed in the table below.

Exhibit	Title	PWS Paragraph	Delivery
A001	Technical Report- PCB Electrical Tests	4.2.2	10 days prior to the end of the period of performance
A002	Technical Report- Impedance test	4.2.2	10 days prior to the end of the period of performance
A003	Technical Report- fabrication and assembly inspection	4.4.2	10 days prior to the end of the period of performance

7.0 Travel

Travel may be required under the performance of this contract and may include:

- First Article Inspection (FAI): To physically inspect and approve the first units produced.
- Production Line Audits: To periodically assess the vendor's manufacturing processes, quality control systems, and handling of government-furnished property or information.

The government may require its technical representatives to be physically present to witness acceptance testing.

- 8 GOVERNMENT FURNISHED EQUIPMENT (GFE): The government may provide Electronic Components necessary for the Contractor to assemble PCBAs. Any components provided by the Government to the contractor shall be identified in a Government Furnished Material (GFM) list, which is a subset of the production Bill of Materials (BOM) required to complete fabrication and assembly. The GFM and BOM may be consolidated into a single list (BOM) so long as quantities of GFM are sufficiently annotated within the BOM.

10 REQUIREMENTS

10.1 **GENERAL REQUIREMENTS:** The contractors shall produce/procure the products described within Section 3. 1, based on the required specification provided by the Government through individual BPA calls. The contractor shall have in-house capabilities for either PCB Fabrication or Assembly, or both.

10.2 **PCB FABRICATION REQUIREMENT:** The contractor shall produce requested quantities of PCBs in accordance with fabrication requirements and specification documents for board thickness, layer count, surface finish, trace tolerance, and any other pertinent fabrication specifications.

10.2.1 The contractor shall fabricate PCBs in accordance with specified IPC-A-600 classification; if IPC Classification is not specified in the provided Technical Data Package (TDP), the contractor shall fabricate PCBs to IPC Class 3 requirements. 4.2.2 The contractor shall provide technical reports, as required, documenting electrical continuity test results, impedance matching test results, and/or results of Quality Control inspections.

10.2.2 Unless otherwise specified in the TDP, the contractor has control over penalization strategies, fiducial markings, and excess materials needed to account for production fall out.

10.2.3 If the contractor does not have internal PCB Fabrication capabilities, it is acceptable to sub-contract PCB Fabrication to another vendor. Any sub-contractors utilized shall be approved by the Government in advance of the initiation of PCB fabrication activities. PCB procurement is payable upon contractor evidence of PO placement, evidence of receipt by contractor, and evidence of acceptable PCB performance (circuit continuity or other test strategy).

10.2.4 The contractor shall return all excess PCB materials to the government after completion of production activities.

10.3 **COMPONENT PROCUREMENT REQUIREMENT:** The contractor shall procure sufficient materials and components, as identified on the BOM, to enable the assembly of requested quantities of PCBAs/CCAs. Any components that will be consigned to the contractor by the Government will be identified on the BOM.

10.3.1 The contractor shall review suitability of (GFM) and components and request any additional information, materials, and/or components within 5 days of receipt of the GFM

- 10.3.2 The contractor shall only use authorized distributors or (OEMs) to source electronic components. Any alternative sources, like third party distributors or brokers, shall be approved by the Government prior to components being procured.
- 10.3.3 The contractor shall rapidly execute component orders, to ensure supply chain disruptions can be immediately accounted for and lead times can be minimized. The contractor shall provide evidence of orders being placed and received. For component and material orders that are excessively costly, partial payments and/or early payments can be negotiated prior to award of the order, upon contractor request. Component procurement is payable upon contractor evidence of PO placement and evidence of receipt by contractor; project inventory shall be transferred to Govt if not used.
- 10.3.4 The contractor shall identify suitable alternates to components that are not immediately available, or that have prohibitively long lead times. Any alternate components shall be approved by the Government technical subject matter experts (SMEs) prior to procurement of the components.
- 10.3.5 Note: The Government may request some additional spares of critical components be procured, to support rework due to post-production test and validation activities; these spares will be identified in the TDP and BOM, when applicable.
- 10.3.6 Unless quantities of components are deliberately specified or limited in the TDP or BOM, the contractor has control over quantities of sufficient spares needed to accommodate losses possible in their PCB Assembly processes. Any alternate components shall be approved by the Government SMEs prior to procurement of the components.
- 10.3.7 The contractor shall return all excess materials and components to the Government after completion of production activities, unless an agreement exists to maintain inventory in contractor inventory in anticipation of follow-on orders.
- 10.3.8 PCB ASSEMBLY REQUIREMENT: The contractor shall assemble requested quantities of PCBAs (also referred to as Circuit Card Assemblies (CCAs)) in accordance with layout specifications and assembly requirement documents for solder selection, component placement, bill of materials, stencil design, and any other pertinent assembly specifications.
- 10.3.9 The contractor shall assemble PCBAs in accordance with specified J-STD-001 and IPC-A-610 classification; if IPC Classification is not specified in the provided TDP, the contract shall assemble PCBAs to IPC Class 2 requirements.

10.3.11 The contractor shall provide technical reports, as required, documenting electrical continuity test results, and/or results of Quality Control inspections performed in accordance with specified IPC-A-610 inspections. (CDRL A003) Unless otherwise specified in the TDP, the contractor shall use Tin-Lead (Sn-Pb) solder in all applicable assembly process.

10.3.12 Note: Some components may be fabricated with Lead-free solder. Unless specified in the TDP, these components shall not be reballed. Hybrid soldering processes are acceptable in the attachment of these components.

11 SERVICE SUMMARY

PWS Paragraph	Performance Objective	Performance Threshold
10.3.8	Fabricate PCBs according to specifications.	All PCBs fabricated to IPC Class 3 (or as specified) with documented evidence of circuit continuity.
10.3.2	Procure all necessary components for assembly.	All components procured from authorized distributors or other Government -approved sources.
10.3.9	Assemble PCBAs according to specifications.	All PCBAs assembled to IPC Class 2 (or as specified) with documented evidence of QC inspections.

12 CDRL SUMMARY

12.1 CDRL Information

12.1.1 Contract data shall be delivered electronically IAW the corresponding requirement for each Data Item listed in the table below. The contractor shall prepare and deliver data to the Government in contractor format except where specific forms are required. Data shall be formatted IAW the requirements specified for each Data Item listed in the table below.

12.1.2 The contractor shall ensure all technical data, including drafts and working papers, are marked and controlled IAW appropriate distribution statements and export control warning notices.

Exhibit	Title	PWS Paragraph	Delivery
A001	Technical Report - PCB Electrical Tests	4.2.2	10 days prior to the end of the period of performance.
A002	Technical Report – Impedance Test	4.2.2	10 days prior to the end of the period of performance.
A003	Technical Report – Fabrication and Assembly Inspections	4.4.2	10 days prior to the end of the period of performance.

13 PLACE/ PERIOD OF PERFORMANCE

13.1 Place of performance

The place of performance shall be the contractors facility.

10.2 Period Of Performance

The period of performance shall be 60 Months, beginning the date of BPA acceptance.

14 PROGRAM MANAGEMENT

14.1 Data Rights Management

All data rights under this contract shall be managed IAW applicable data rights clauses outlined in the FAR and DFARS. The U.S. Government shall retain unlimited data and computer software rights to all items delivered under this contract. Any data delivered to the contractor shall remain property of the US Government.

14.2 Government Furnished Property

The government may provide Electronic components necessary for the contractor to assemble PCBA's. Any components that will be provided by the government to the contractor shall be identified in a (GFM) list. The contractor shall review the suitability of GFM and request and additional information or materials within 5 days of receipt.

14.3 Quality Control Plan

The contractor shall establish a complete Quality Control Plan (QCP) to ensure the requirements of this contract are provided as specified.

14.4 Quality Assurance

The Government will perform assessments of the contractor's performance. The Government may inspect each task as completed or increase the number of quality control inspections if deemed appropriate.

14.5 Conflicts of Interest

The Contractor shall not assign to the contract or task order performance, any person, if employing that person would create a conflict of interest under any law, regulation, or policy of the United States Government.

14.6 Government Observation

During the PoP, Government personnel other than the Contracting Officer (CO) may, from time-to-time, observe contractor operations. However, these personnel may not interfere with contractor performance.

APPENDIX A – NAME OF APPENDIX